

Virtual Machine Provisioning with Microsoft Azure

Submitted By : Dhona Nancy A

Submitted To : Tech Vedhu Team

Institution : Panimalar Engineering College

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INTRODUCTION AND OBJECTIVES

Cloud computing has become an essential technology for deploying, managing, and scaling applications without relying on physical infrastructure. Microsoft Azure is one of the leading cloud platforms that provides Infrastructure-as-a-Service (IaaS) solutions, enabling users to create and manage virtual machines, networking resources, storage, and monitoring services through a secure cloud environment.

In this project, an Ubuntu Server 24.04 LTS Virtual Machine was provisioned and configured using Microsoft Azure. Secure remote access was established through SSH authentication, allowing administrative access to the virtual machine. Basic Linux administration tasks were performed to verify system functionality, and an Apache web server was installed to host a web page that could be accessed through the VM's public IP address.

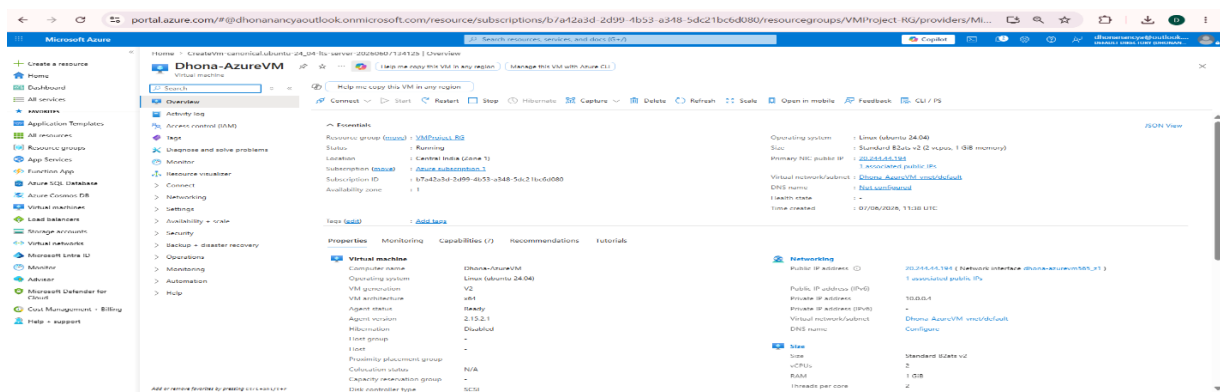
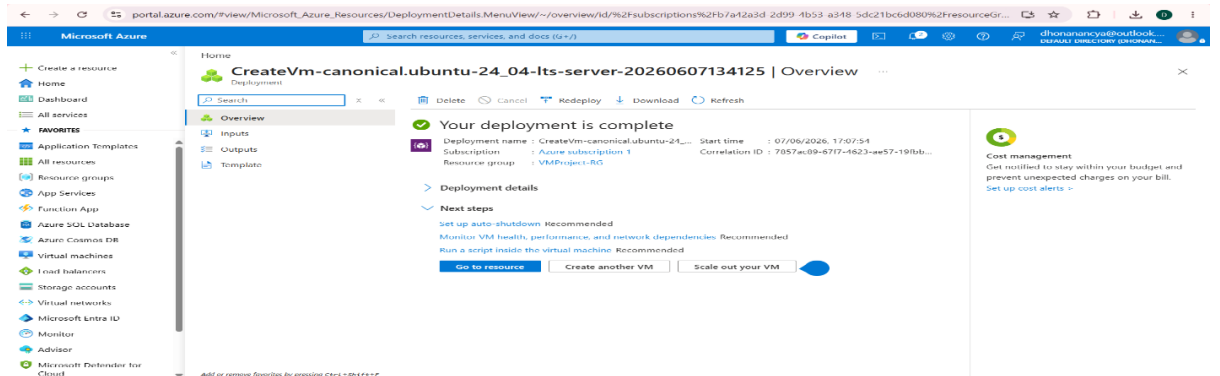
The primary objectives of this project were:

- To provision and configure an Ubuntu Virtual Machine in Microsoft Azure.
- To establish secure remote access using SSH public key authentication.
- To configure networking and security settings required for remote connectivity and web hosting.
- To perform basic Linux system administration and resource verification.
- To install and manage the Apache web server on the virtual machine.
- To monitor virtual machine performance using Azure Metrics.
- To understand the concept of Auto Scaling using Azure Virtual Machine Scale Sets (VMSS).
- To gain practical experience in cloud infrastructure deployment and management using Microsoft Azure.

TECHNOLOGIES USED

The following technologies were used throughout the project to create, manage, monitor, and access the virtual machine hosted in Microsoft Azure. The table below outlines the primary cloud, operating system, network, and application components leveraged throughout this deployment:

TECHNOLOGY	PURPOSE
Microsoft Azure	Core Enterprise Cloud Platform providing Infrastructure-as-a-Service (IaaS).
Ubuntu Server 24.04 LTS	Open-source, enterprise Linux distribution used as the host Operating System.
Secure Shell (SSH)	Network protocol used for encrypted cryptographic remote terminal administration.
Apache2 HTTP Server	Web server software deployed to process global public HTTP requests.
Azure Monitor Metrics	Integrated platform telemetrics used to monitor real-time infrastructure health.
Virtual Machine Scale Sets	Framework concept used to design elastic automatic horizontal resource scaling.



VIRTUAL MACHINE PROVISIONING

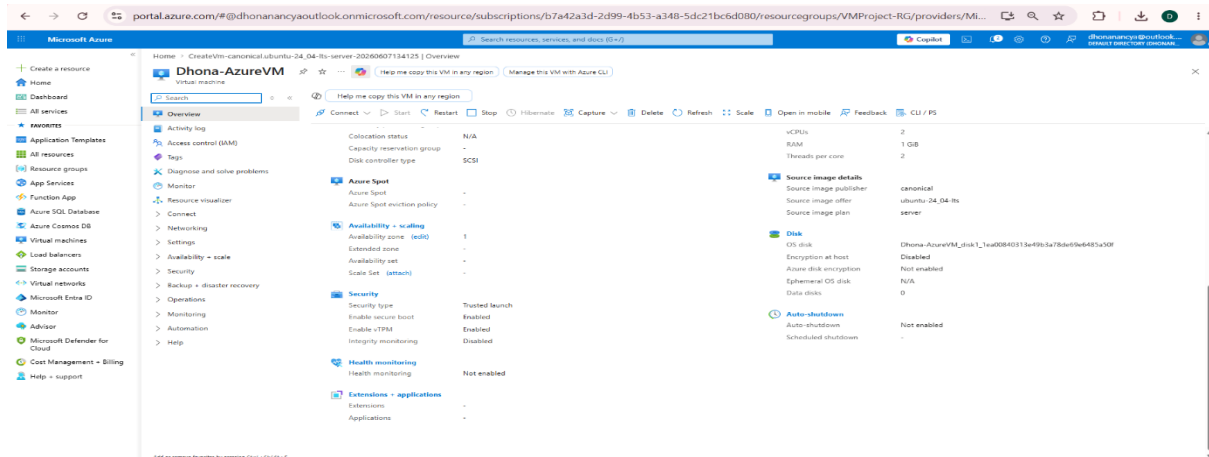
The virtual machine was successfully deployed in Microsoft Azure using Ubuntu Server 24.04 LTS. Resource groups, networking, authentication, and security settings were configured during the provisioning process.

Resource Details

- Resource Group: VMProject-RG
- VM Name: Dhona-AzureVM
- Region: Central India
- OS: Ubuntu Server 24.04 LTS
- Authentication: SSH Public Key

Deployment Process

1. Logged into Azure Portal.
2. Created Resource Group.
3. Selected Ubuntu Server image.
4. Configured VM specifications.
5. Enabled SSH access.
6. Created and deployed VM.



Status : Running
Location : Central India (Zone 1)
Subscription ([move](#)) : [Azure subscription 1](#)

Size : Standard B2ats v2 (2 vcpus, 1 GiB memory)
Primary NIC public IP : [20.244.44.194](#)
[1 associated public IPs](#)

SSH ACCESS AND LINUX ADMINISTRATION

The VM was accessed securely through SSH using a generated SSH key pair.

Commands executed:

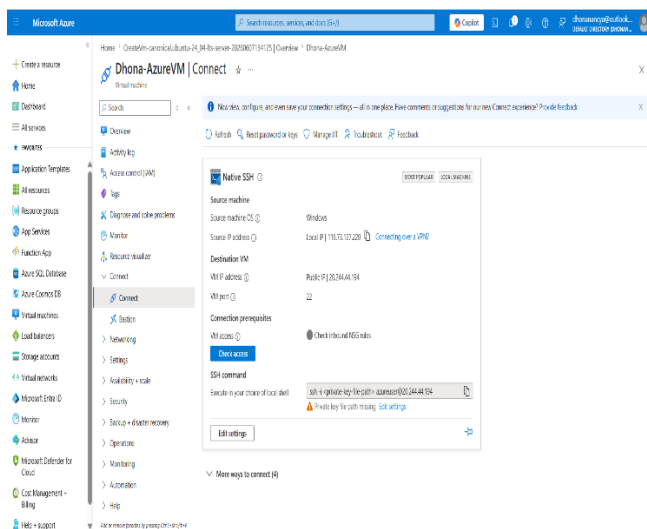
sudo apt update : Synchronizes the local package index files with their remote repositories over the internet to discover updated system versions.

sudo apt upgrade -y : Downloads and installs the latest verified stability and security patches for all existing packages currently running on the server.

uname -a : Prints complete system information to the terminal, including the exact Linux kernel version, host machine name, and hardware operating structure.

free -h : Displays the exact amount of free and used physical memory (RAM) and swap space within the cloud server in a human-readable format.

lscpu : Gathers and lists highly detailed information regarding the CPU architecture, core counts, processor speeds, and cache dimensions from the underlying virtualized hardware.



```
Copyright (C) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements: https://aka.ms/PSWindows
PS C:\Users\A.DHONANANCY> cd "C:\Users\A.DHONANANCY\Downloads\Azure Project"
PS C:\Users\A.DHONANANCY\Downloads\Azure Project> dir

Directory: C:\Users\A.DHONANANCY\Downloads\Azure Project

Mode                LastWriteTime         Length Name
----                -
d-----          07-06-2026      18:07          Azure VM Project ( Screenshot )
-a-----          07-06-2026      17:07          2498 Dhona-AzureVM_key.pem

PS C:\Users\A.DHONANANCY\Downloads\Azure Project> ssh -i Dhona-AzureVM_key.pem azureuser@20.244.44.194
The authenticity of host '20.244.44.194 (20.244.44.194)' can't be established.
ED25519 key fingerprint is SHA256:17Xmapth1A2n+HkzyWR1TSRFPcU1170ESMe1zVC3G9.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.244.44.194' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1018-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sun Jun  7 12:58:20 UTC 2026

System load:  0.0               Processes:    118
Usage of /:   5.8% of 28.02GB   Users logged in:  0
Memory usage: 33%              IPv6 address for eth0: 10.0.0.0
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@Dhona-AzureVM:~$
```

```
azureuser@Dhona-AzureVM:~$ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]

All packages are up to date.
azureuser@Dhona-AzureVM:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
azureuser@Dhona-AzureVM:~$ uname -a
Linux Dhona-AzureVM 6.17.0-1018-azure #18~24.04.1-Ubuntu SMP Thu May 28 16:39:11 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
azureuser@Dhona-AzureVM:~$
```

SYSTEM INFORMATION

After successfully connecting to the Ubuntu Virtual Machine through SSH, several Linux commands were executed to verify the system configuration and resource availability. These commands provided detailed information about the operating system, memory allocation, processor specifications, and overall system architecture. Monitoring these resources helps administrators understand the VM's performance and ensures that sufficient computing resources are available for applications and services. The collected information also confirms that the virtual machine was deployed correctly and is functioning as expected.

RAM Information

The `free -h` command was used to display the available and utilized memory in a human-readable format. This information helps in understanding the memory consumption of the system and assists in performance monitoring and capacity planning.

```
azureuser@Dhona-AzureVM:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           846Mi       395Mi       137Mi         4.0Mi        455Mi        451Mi
Swap:            0B           0B           0B
```

CPU Information

The `lscpu` command was executed to obtain detailed processor information, including architecture, CPU cores, virtualization support, and processing capabilities. These details are useful for evaluating the computational power of the virtual machine.

```
azureuser@Dhona-AzureVM:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          48 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 2
On-line CPU(s) list:   0,1
Vendor ID:              AuthenticAMD
Model name:             AMD EPYC 7763 64-Core Processor
CPU family:             25
Model:                  1
Thread(s) per core:     2
Core(s) per socket:     1
Socket(s):              1
Stepping:               1
BogoMIPS:               4899.87
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt
                        pdpe1gb rdtscp lm constant_tsc rep_good nopl tsc_reliable nonstop_tsc cpuid extd_apicid aperfmperf tsc_known_freq pni pclmulqd
                        q sse3_fma cx16 pcid sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy cr8_legacy abm sse4a m
                        isalignsse 3dnowprefetch osvw topoext vmcall fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni
                        xsaveopt xsavec xgetbv1 xsaves user_shstk clzero xsaveerptr rdpru arat unip vaes vpclmulqdq rdpid fsrm
Virtualization features:
Hypervisor vendor:     Microsoft
Virtualization type:    full
Caches (sum of all):
L1d:                   32 KiB (1 instance)
L1i:                   32 KiB (1 instance)
L2:                    512 KiB (1 instance)
L3:                    32 MiB (1 instance)
NUMA:
NUMA node(s):          1
NUMA node0 CPU(s):     0,1
Vulnerabilities:
Gather data sampling:   Not affected
Ghostwrite:             Not affected
Indirect target selection: Not affected
Itlb multihit:         Not affected
L1tf:                  Not affected
Mds:                   Not affected
Meltdown:              Not affected
Mmio stale data:        Not affected
Old microcode:          Not affected
Reg file data sampling: Not affected
Retbleed:              Not affected
Spec rstack overflow:   Vulnerable: Safe RET, no microcode
Spec store bypass:      Vulnerable
Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:             Mitigation; Retpolines; STIBP disabled; RSB filling; PBRSB-eIBRS Not affected; BHI Not affected
Srbds:                 Not affected
Tsx:                   Vulnerable: No microcode
Tsx async abort:        Not affected
Vmscape:               Not affected
azureuser@Dhona-AzureVM:~$
```

APACHE WEB SERVER DEPLOYMENT

To host a web application on the virtual machine, the Apache HTTP Server was installed using Ubuntu package management tools. Apache is one of the most widely used open-source web servers and provides a reliable platform for hosting websites and web applications. After installation, the Apache service was started and configured to launch automatically whenever the virtual machine boots. Service status verification confirmed that the web server was running successfully without any errors.

The successful deployment of Apache demonstrates how cloud-hosted virtual machines can be used as web servers to deliver content over the internet. The default Apache page was accessed through the VM's public IP address, confirming that networking and web server configurations were functioning correctly.

Commands Used :

```
sudo apt install apache2 -y
```

```
sudo systemctl start apache2
```

```
sudo systemctl enable apache2
```

```
sudo systemctl status apache2
```

```
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module ssl.
Enabling module socshandshake.
Enabling module charset.
Enabling module localised-error-pages.
Enabling module other-vhosts-access-log.
Enabling module security.
Enabling module cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36-2.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@Dhona-AzureVM:~$ |
```

```
azureuser@Dhona-AzureVM:~$ sudo systemctl start apache2
azureuser@Dhona-AzureVM:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
azureuser@Dhona-AzureVM:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-06-07 13:04:08 UTC; 2min 31s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 3011 (apache2)
    Tasks: 55 (limit: 988)
   Memory: 5.2M (peak: 5.6M)
      CPU: 47ms
   CGroup: /system.slice/apache2.service
           └─3011 /usr/sbin/apache2 -k start
           └─3014 /usr/sbin/apache2 -k start
           └─3015 /usr/sbin/apache2 -k start

Jun 07 13:04:08 Dhona-AzureVM systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jun 07 13:04:08 Dhona-AzureVM systemd[1]: Started apache2.service - The Apache HTTP Server.
azureuser@Dhona-AzureVM:~$ |
```



Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
    |-- ports.conf
    |-- mods-enabled
        |-- *.conf
    |-- conf-enabled
        |-- *.conf
    |-- sites-enabled
        |-- *.conf
```

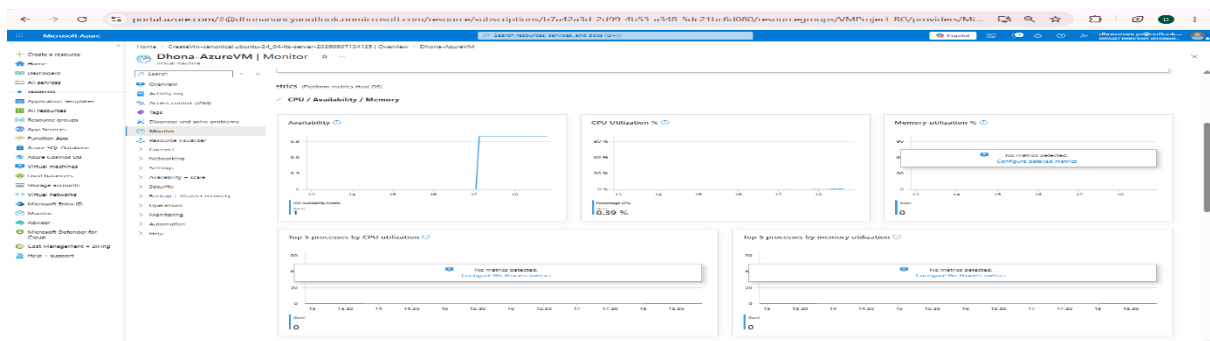
MONITORING

Microsoft Azure provides built-in monitoring tools that help users track the performance and health of virtual machines in real time. Monitoring is important for understanding resource utilization and ensuring that the system operates efficiently. Azure Metrics allows users to visualize performance data through graphs and charts.

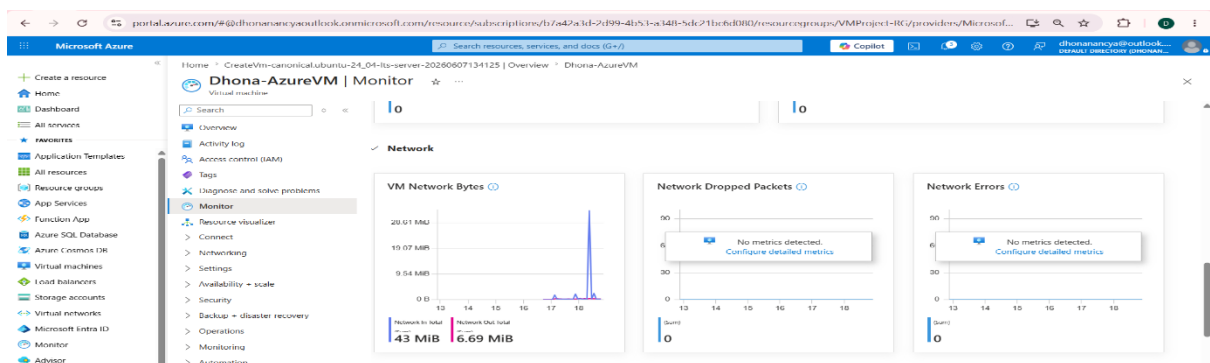
In this project, Azure Monitor was used to observe key performance metrics of the virtual machine. CPU utilization, network traffic, and disk activity were analyzed to evaluate the VM's operational status. These metrics help administrators identify performance trends and maintain system reliability.

Metrics Observed

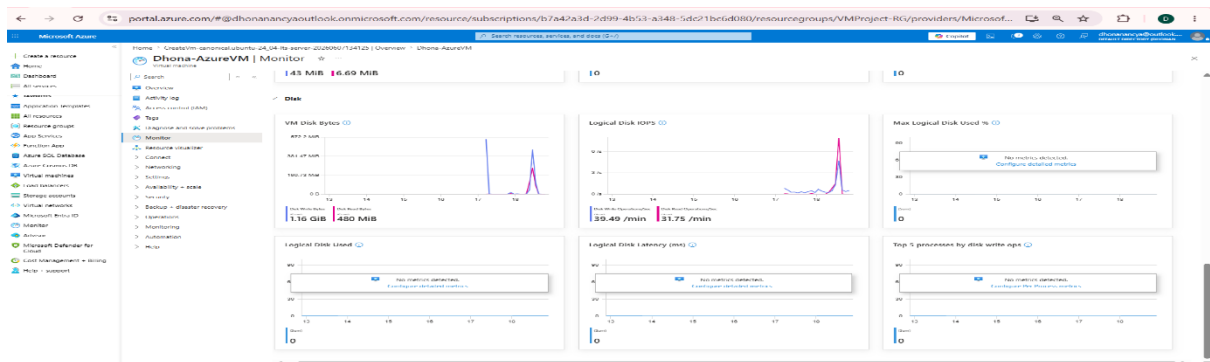
- CPU Utilization



- Network Traffic



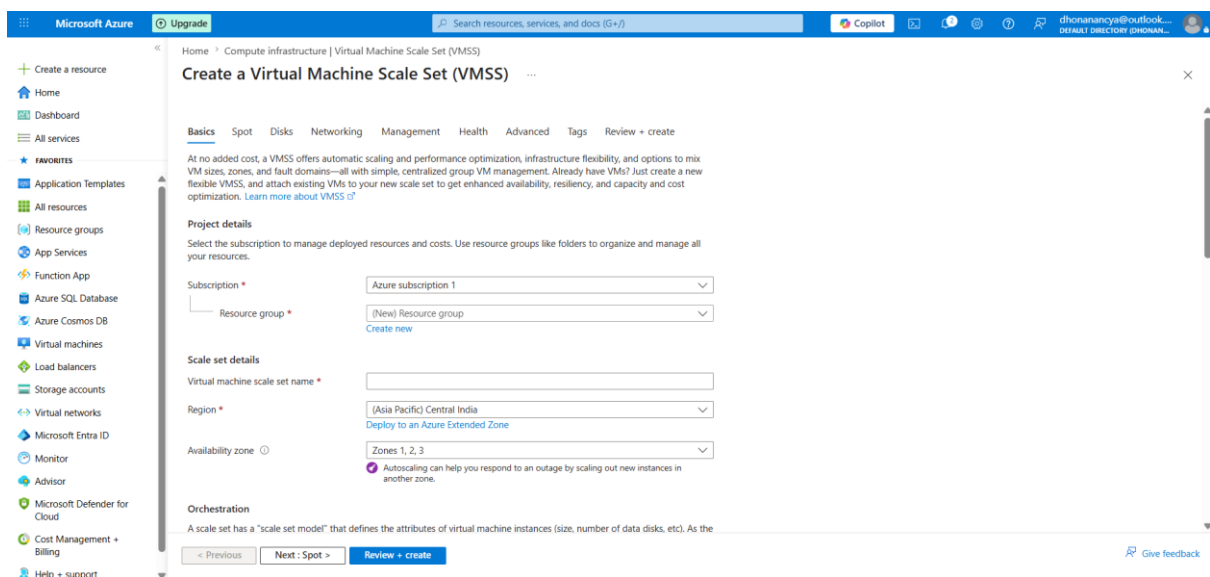
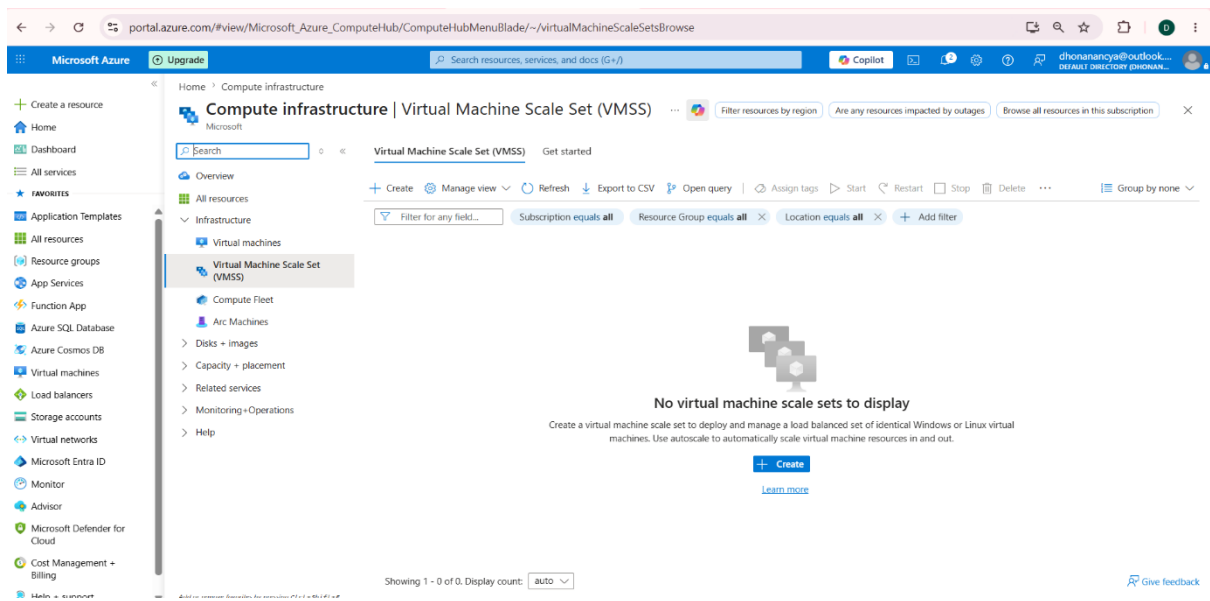
- Disk Read/Write Activity



AUTO SCALING

Auto Scaling is a cloud computing feature that automatically adjusts resources based on workload requirements. Microsoft Azure provides this functionality through Virtual Machine Scale Sets (VMSS), which can automatically increase or decrease the number of virtual machine instances.

This feature helps improve application availability and performance while optimizing resource usage and operational costs. During this project, the VM Scale Sets service was explored to understand Azure's approach to automated scaling and resource management.



RESULTS AND CONCLUSION

Project Results :

The objectives of this project were successfully achieved through the deployment and management of a virtual machine in Microsoft Azure. An Ubuntu Server 24.04 LTS virtual machine was provisioned and configured with the required networking and security settings. Secure remote access was established using SSH public key authentication, enabling safe administration of the virtual machine from a Windows environment.

Several Linux administration commands were executed to verify the operating system, memory, and processor details. An Apache Web Server was successfully installed, configured, and started on the virtual machine. The default Apache web page was accessed through the public IP address, confirming that the web server and networking configurations were functioning correctly.

In addition, Azure Monitor was used to observe key performance metrics such as CPU utilization, network traffic, and disk activity. The Virtual Machine Scale Sets (VMSS) service was also explored to understand Azure's auto-scaling capabilities and resource management features.

Conclusion :

This project provided comprehensive, hands-on experience in cloud infrastructure deployment, modern networking configurations, secure remote administration, and web server optimization within the Microsoft Azure platform. The implementation successfully demonstrated the operational flexibility, rapid deployment capabilities, and inherent scalability of modern Infrastructure-as-a-Service (IaaS) environments.

Through this practical exercise, a foundational understanding of configuring enterprise cloud boundaries, executing administrative Linux terminal commands, interpreting live performance graphs, and designing automated auto-scaling architectures was achieved, bridging the gap between theoretical cloud concepts and live deployment patterns.

Project URL : <https://github.com/dhonanancy/Azure-VM-Provisioning-Project>

GitHub Repository: <https://github.com/dhonanancy/Azure-VM-Provisioning-Project>